

MINI-PROJECT REPORT

Local Telemetry Generation

Windows event logging with Sysmon and Security auditing

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1. Objective

The aim of this mini-project is to set up local logging on a Windows system, generate five distinct security events by hand, and show exactly where the resulting logs are stored. For each event the report highlights the three fields an analyst reads first: the timestamp, the user and the process.

Highlight colour code used in the screenshots:

Colour	Meaning
Red box	Timestamp fields (UtcTime, TimeCreated, Logged)
Blue box	User fields (User, TargetUserName, SubjectUserName, SamAccountName)
Green box	Process fields (Image, ProcessId, CommandLine, ProcessName)
Orange box	Other fields that help interpret the event (Event ID, status codes, parent process, IP address)

2. Environment and tools

Item	Detail
Operating system	Windows, build 10.0.26100 (as shown in the Sysmon FileVersion field)
Host name	Dnyaneshwari
Deployment	Local Windows machine
Logging utility	Sysmon (System Monitor) v15.22, installed with default configuration
Built-in auditing	Windows Security auditing configured with auditpol (Logon, User Account Management)
Viewing tools	Event Viewer (Friendly and XML views) and Windows PowerShell
Evidence folder	C:\LocalTelemetry\Screenshots

3. Configuration

3.1 Enabling Windows audit policy

Two audit subcategories were enabled for both success and failure so that the Security log records logons and account changes. The settings were then read back to confirm they applied.

```
auditpol /set /subcategory:"Logon" /success:enable /failure:enable
auditpol /set /subcategory:"User Account Management" /success:enable
/failure:enable
```

```

PS C:\WINDOWS\system32> auditpol /set /subcategory:"Logon" /success:enable /failure:enable
The command was successfully executed.
PS C:\WINDOWS\system32> auditpol /set /subcategory:"User Account Management" /success:enable /failure:enable
The command was successfully executed.
PS C:\WINDOWS\system32> auditpol /get /subcategory:"Logon"
System audit policy
Category/Subcategory          Setting
Logon/Logoff
  Logon                        Success and Failure
PS C:\WINDOWS\system32> auditpol /get /subcategory:"User Account Management"
System audit policy
Category/Subcategory          Setting
Account Management
  User Account Management     Success and Failure
PS C:\WINDOWS\system32>

```

Figure 1: auditpol commands enabling and verifying Logon and User Account Management auditing (Success and Failure).

3.2 Installing Sysmon

The Sysmon package was extracted to C:\Sysmon, which contains the Sysmon.exe, Sysmon64.exe and Sysmon64a.exe binaries and the licence file.

```

PS C:\WINDOWS\system32> cd C:\Sysmon
PS C:\Sysmon> dir

Directory: C:\Sysmon

Mode                LastWriteTime         Length Name
----                -
-a----           10/4/2026  11:34 AM             7490 Eula.txt
-a----           10/4/2026  11:34 AM        6162240 Sysmon.exe
-a----           10/4/2026  11:34 AM       3203368 Sysmon64.exe
-a----           10/4/2026  11:34 AM       3138824 Sysmon64a.exe

```

Figure 2: Contents of C:\Sysmon after extracting the Sysmon package.

Sysmon was then installed as a service and driver with the 64-bit binary, and the service state was checked.

```

.\Sysmon64.exe -accepteula -i
Get-Service Sysmon64

```

```

PS C:\Sysmon> .\Sysmon64.exe -accepteula -i

System Monitor v15.22 - System activity monitor
By Mark Russinovich and Thomas Garnier
Copyright (C) 2014-2026 Microsoft Corporation
Using libxml2. libxml2 is Copyright (C) 1998-2012 Daniel Veillard. All Rights Reserved.
Sysinternals - www.sysinternals.com

Sysmon64 installed.
SysmonDrv installed.
Starting SysmonDrv.
SysmonDrv started.
Starting Sysmon64..
Sysmon64 started.
PS C:\Sysmon> Get-Service Sysmon64

Status      Name             DisplayName
-----
Running     Sysmon64         Sysmon64

PS C:\Sysmon>

```

Figure 3: Sysmon v15.22 installed; SysmonDrv and Sysmon64 started, and Get-Service shows Sysmon64 Running.

Because no configuration file was supplied, Sysmon ran with its default settings. This is why the RuleName field shows "-" in the events below.

4. Events generated

Five distinct events were generated on 4 October 2026. Sysmon stores times in UTC, while Event Viewer shows local time (IST, UTC+5:30), so each event is listed in both.

#	Event	Log	Event ID	Time (UTC)	Time (local, IST)
1	Process creation (Notepad)	Sysmon	1	06:11:43	11:41:43 AM
2	New local user created	Security	4720 / 4722	06:39:29 (4720) 06:13:37 (4722)	12:09:29 PM (4720) 11:43:37 AM (4722)
3	Failed login	Security	4625	06:20:05	11:50:05 AM
4	PowerShell command	Sysmon	1	06:29:44	11:59:44 AM
5	Command Prompt command	Sysmon	1	06:58:37	12:28:37 PM

5. Event details

5.1 Event 1: Process creation (Sysmon Event ID 1)

Notepad was launched to produce a baseline process-creation event. Sysmon logs every new process with its full image path, process ID and a unique process GUID.

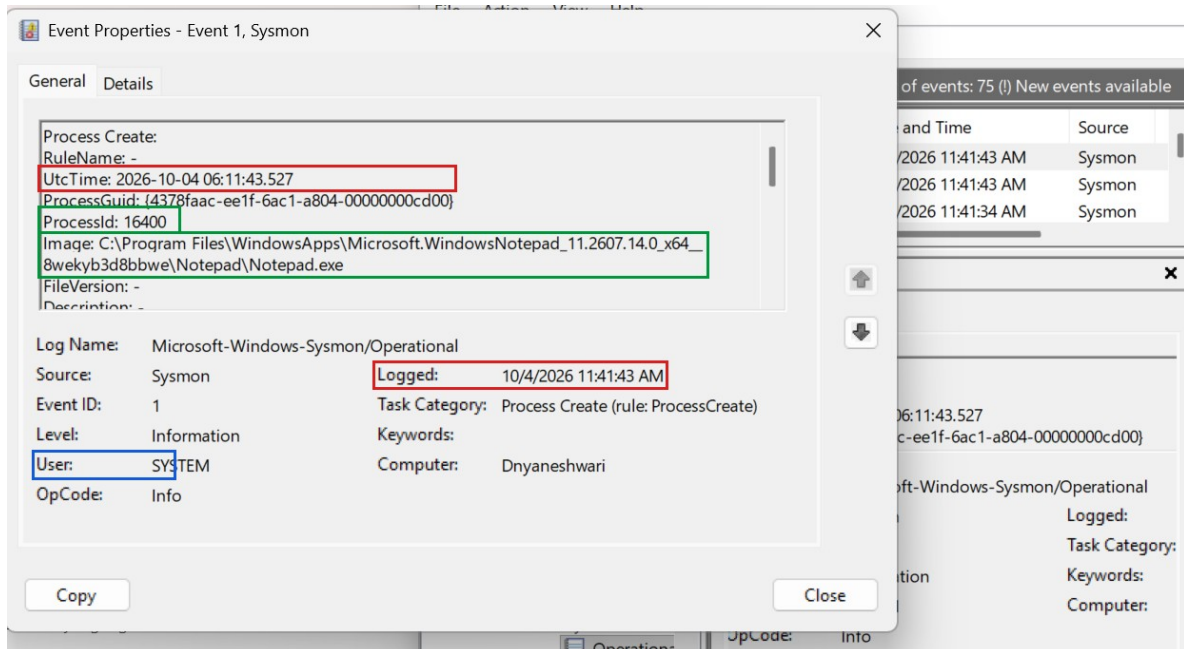


Figure 4: Sysmon Event ID 1 for Notepad.exe, with timestamp, process and user fields highlighted.

Category	Field	Value
Timestamp	UtcTime / Logged	2026-10-04 06:11:43.527 / 10/4/2026 11:41:43 AM
Process	Image	C:\Program Files\WindowsApps\Microsoft.WindowsNotepad_11.2607.14.0_x64__8wekyb3d8bbwe\Notepad\Notepad.exe
Process	ProcessId	16400
User	User (event record)	SYSTEM, the account Sysmon writes the record under
Detail	Log / Event ID	Microsoft-Windows-Sysmon/Operational / 1 (Process Create)

5.2 Event 2: New local user created (Security Event ID 4720)

A new local account was added from an elevated PowerShell prompt, and its properties were then listed to confirm it exists. The test password is a throwaway lab value for an account that exists only for this exercise.

```
net user TelemetryTest P@ssw0rd123! /add
net user TelemetryTest
```

```

PS C:\Sysmon> net user TelemetryTest P@ssw0rd123! /add
The command completed successfully.

PS C:\Sysmon> net user TelemetryTest
User name                TelemetryTest
Full Name
Comment
User's comment
Country/region code      000 (System Default)
Account active            Yes
Account expires           Never

Password last set        10/4/2026 11:43:38 AM
Password expires         11/15/2026 11:43:38 AM
Password changeable      10/4/2026 11:43:38 AM
Password required         Yes
User may change password Yes

Workstations allowed     All
Logon script
User profile
Home directory
Last logon                Never

Logon hours allowed      All

Local Group Memberships  *Users
Global Group memberships *None
The command completed successfully.

```

Figure 5: Account TelemetryTest created and verified with net user (active, member of the local Users group, last logon: Never).

Windows records account management in the Security log. Event 4722 (account enabled) was captured for TelemetryTest, and Event 4720 (account created) was captured for a second account, TelemetryTest2.



Figure 6: Security Event ID 4722 (user account enabled) for TelemetryTest.

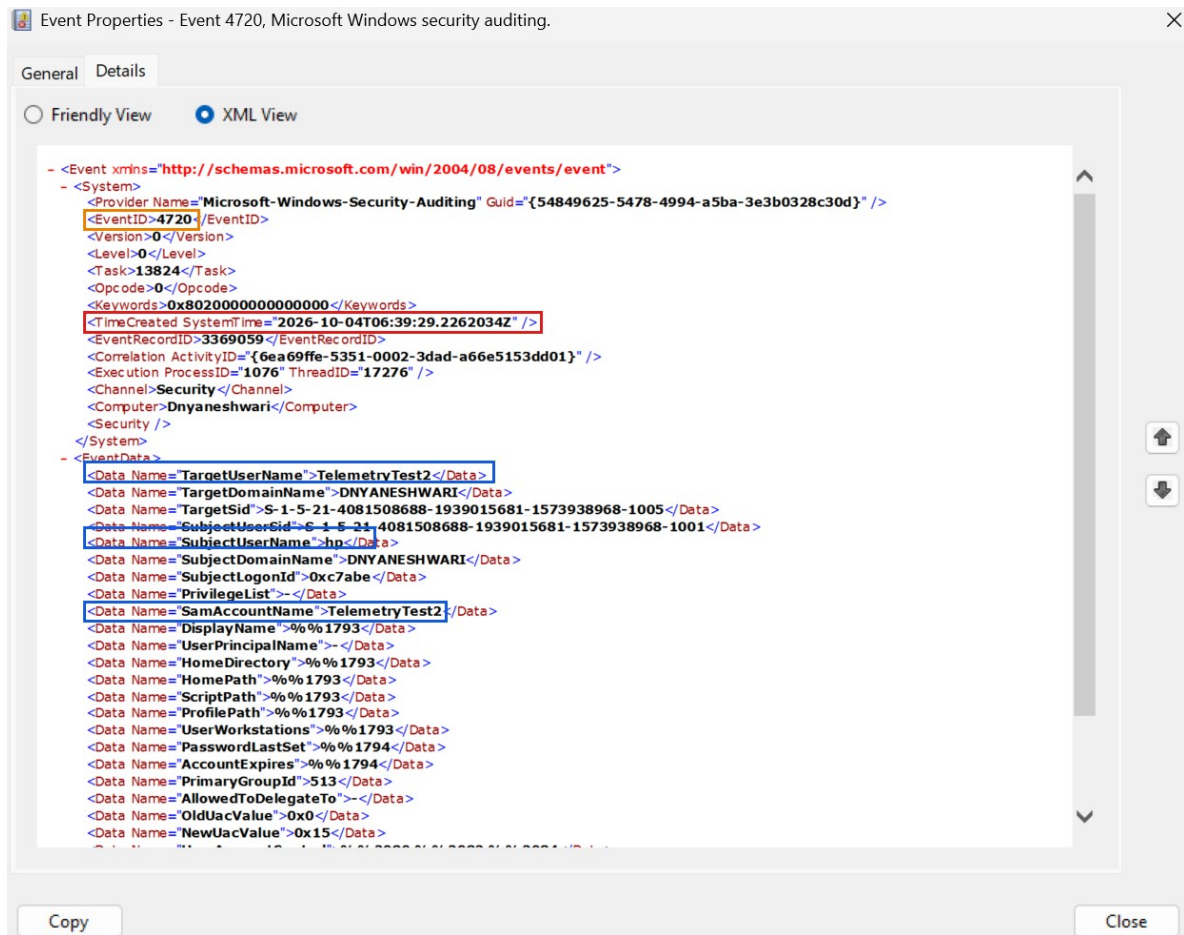


Figure 7: Security Event ID 4720 (user account created) for TelemetryTest2, with timestamp and user fields highlighted.

Category	Field	Value (Event 4720)
Timestamp	TimeCreated SystemTime	2026-10-04T06:39:29.2262034Z
User	TargetUserName / SamAccountName	TelemetryTest2 (the new account)
User	SubjectUserName	hp (the account that created it), domain DNYANESHWARI
Detail	SubjectLogonId	0xc7abe (the logon session that performed the action)
Detail	EventRecordID / Channel	3369059 / Security
Process	Process	Not recorded in 4720; the acting session is identified by SubjectLogonId

5.3 Event 3: Failed login (Security Event ID 4625)

A logon attempt for the account TelemetryTest was made with an incorrect password. Windows logged the failure as Event 4625.

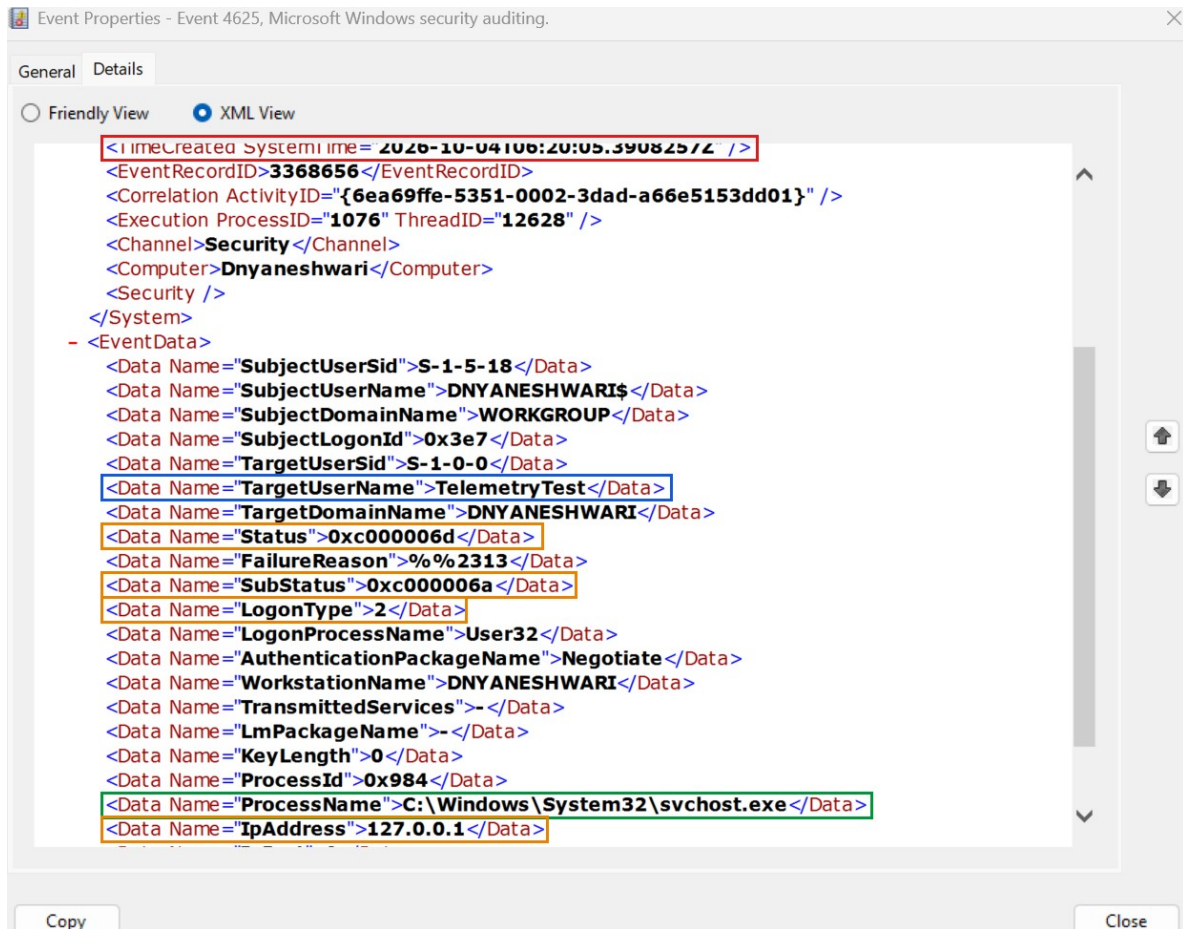


Figure 8: Security Event ID 4625, with the target user, status codes and process highlighted.

Category	Field	Value
Timestamp	TimeCreated SystemTime	2026-10-04T06:20:05Z (11:50:05 AM IST)
User	TargetUserName	TelemetryTest (the account the logon was attempted for)
Detail	Status	0xc000006d (logon failure)
Detail	SubStatus	0xc000006a (user name valid, password incorrect)
Detail	LogonType	2 (interactive logon)
Process	ProcessName	C:\Windows\System32\svchost.exe
Detail	IpAddress	127.0.0.1 (local machine)
Detail	SubjectUserName	DNYANESHWARI\$ (the machine account that submitted the request)

Reading the event: the combination of Status 0xc000006d and SubStatus 0xc000006a tells an analyst the account name was real but the password was wrong. If the user name did not exist, the SubStatus would differ.

5.4 Event 4: PowerShell command (Sysmon Event ID 1)

A PowerShell command was run from the C:\Sysmon directory. Sysmon logged the new powershell.exe process together with its complete command line.

```
powershell.exe -NoProfile -Command "Write-Output Event4-Test"
```

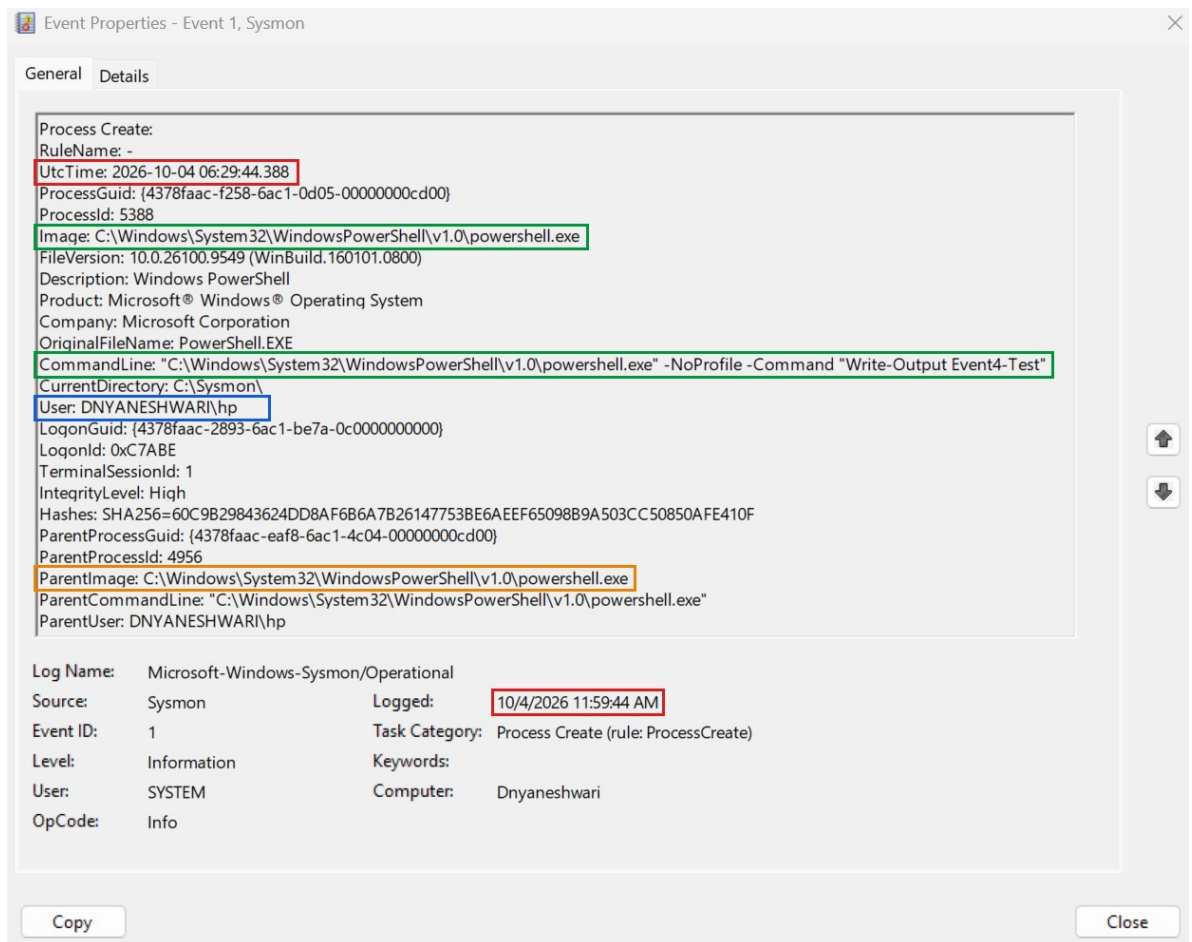


Figure 9: Sysmon Event ID 1 for powershell.exe, with timestamp, process, command line and user highlighted.

Category	Field	Value
Timestamp	UtcTime / Logged	2026-10-04 06:29:44.388 / 10/4/2026 11:59:44 AM
Process	Image	C:\Windows\System32\WindowsPowerShell\v1.0\powershell.exe
Process	ProcessId	5388
Process	CommandLine	"...\powershell.exe" -NoProfile -Command "Write-Output Event4-Test"
User	User	DNYANESHWARI\hp
Detail	ParentImage / ParentProcessId	powershell.exe / 4956
Detail	IntegrityLevel / LogonId	High / 0xC7ABE

Category	Field	Value
Detail	Hashes	SHA256=60C9B29843624DD8AF6B6A7B26147753BE6A EEF65098B9A503CC50850AFE410F

5.5 Event 5: Command Prompt command (Sysmon Event ID 1)

A Command Prompt instance was started from PowerShell to run a single echo command. Sysmon recorded cmd.exe with PowerShell as its parent.

```
cmd.exe /c "echo Event5-Test"
```

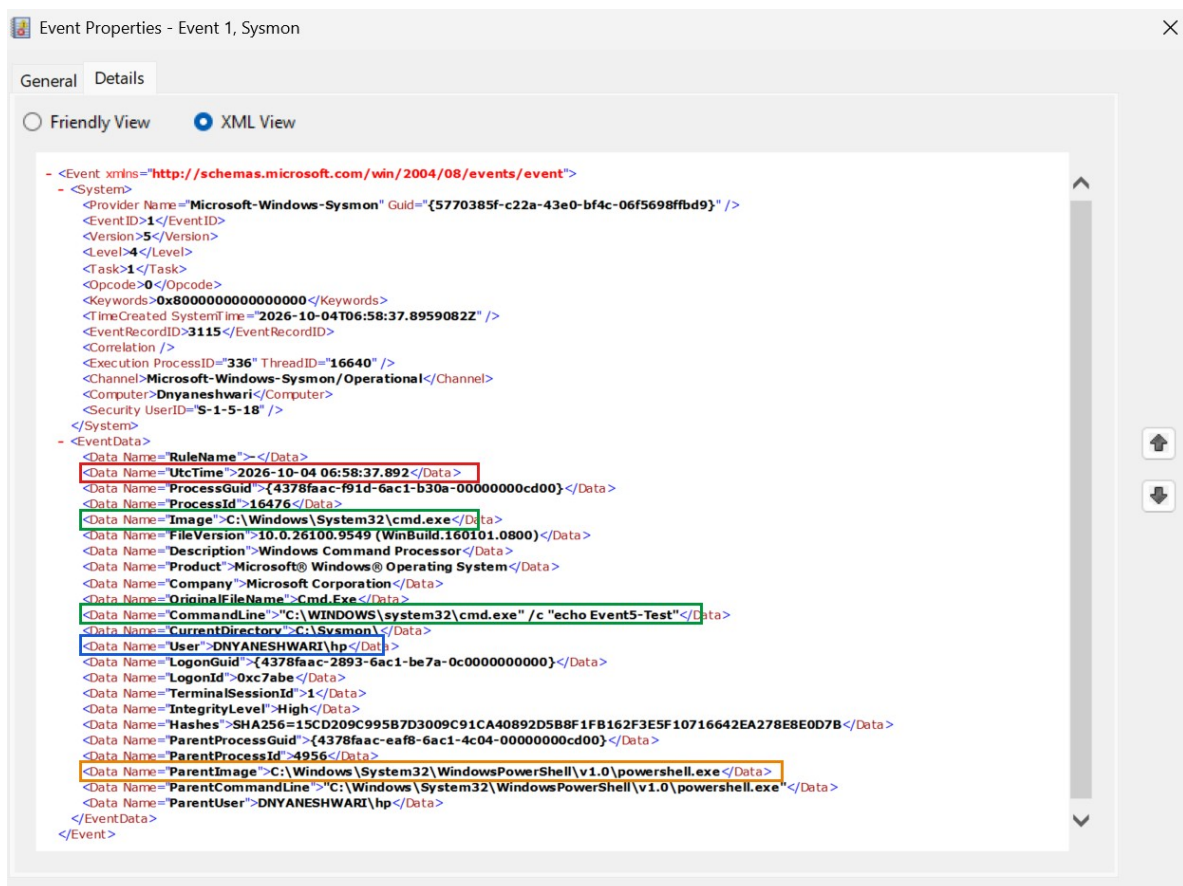


Figure 10: Sysmon Event ID 1 (XML view) for cmd.exe, with timestamp, process, user and parent process highlighted.

Category	Field	Value
Timestamp	UtcTime	2026-10-04 06:58:37.892
Process	Image	C:\Windows\System32\cmd.exe
Process	ProcessId	16476
Process	CommandLine	"C:\WINDOWS\system32\cmd.exe" /c "echo Event5-Test"
User	User	DNYANESHWARI\hp

Category	Field	Value
Detail	ParentImage / ParentProcessId	powershell.exe / 4956
Detail	EventRecordID	3115
Detail	Hashes	SHA256=15CD209C995B7D3009C91CA40892D5B8F1FB162F3E5F10716642EA278E8E0D7B

6. Where the logs are stored

Both logs are standard Windows event log files (.evtx) in the same folder. They can be opened in Event Viewer or read directly from disk.

Log	File path	Event Viewer location
Security	C:\Windows\System32\winevt\Logs\Security.evtx	Windows Logs > Security
Sysmon	C:\Windows\System32\winevt\Logs\Microsoft-Windows-Sysmon%4Operational.evtx	Applications and Services Logs > Microsoft > Windows > Sysmon > Operational

```

PS C:\Sysmon> Get-ChildItem "C:\Windows\System32\winevt\Logs\*Sysmon*"

Directory: C:\Windows\System32\winevt\Logs

Mode                LastWriteTime         Length Name
----                -
-a----           10/4/2026 12:31 PM      4263936 Microsoft-Windows-Sysmon%4Operational.evtx

PS C:\Sysmon> Get-Item "C:\Windows\System32\winevt\Logs\Security.evtx"

Directory: C:\Windows\System32\winevt\Logs

Mode                LastWriteTime         Length Name
----                -
-a----           10/4/2026 12:30 PM      20975616 Security.evtx

```

Figure 11: PowerShell listing of both log files in C:\Windows\System32\winevt\Logs. Sizes: Sysmon 4,263,936 bytes; Security 20,975,616 bytes.

The Sysmon log properties in Event Viewer confirm the log path, a maximum size of 65,536 KB and the overwrite-as-needed retention setting.

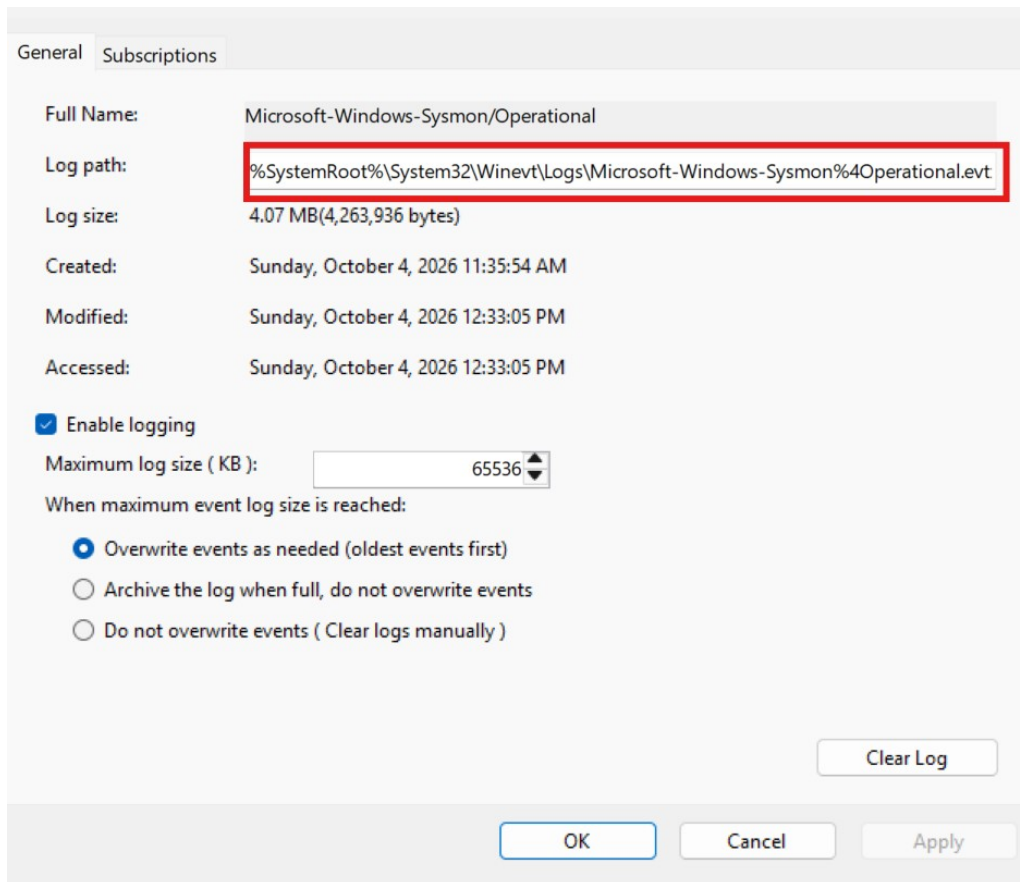


Figure 12: Event Viewer log properties for Microsoft-Windows-Sysmon/Operational, with the log path highlighted.

7. Analysis

7.1 What each event tells an analyst

#	Event	Detection value	Related MITRE ATT&CK technique
1	Process creation	Shows what ran, from which path, and under which session; the baseline for process-based hunting.	Baseline telemetry (no single technique)
2	User created (4720, 4722)	Unexpected new accounts are a common persistence step; the Subject fields show who created it.	T1136.001 Create Account: Local Account
3	Failed login (4625)	A single failure is normal. Repeated failures for one or many accounts point to password guessing.	T1110 Brute Force (when repeated)

#	Event	Detection value	Related MITRE ATT&CK technique
4	PowerShell (Sysmon 1)	The full command line exposes what was executed, including encoded or obfuscated commands.	T1059.001 PowerShell
5	cmd.exe (Sysmon 1)	The parent process shows how the shell was launched, here PowerShell spawning cmd.exe.	T1059.003 Windows Command Shell

7.2 Correlating events across logs

The logon session ID links activity in the two logs. The 4720 event shows SubjectLogonId 0xc7abe, and the PowerShell and cmd.exe events in Sysmon show LogonId 0xC7ABE for the same user, hp. This shows that the account creation and the shell activity came from one logon session, which is the kind of link an analyst uses to reconstruct a sequence of actions.

The two PowerShell-related events also share a parent: both show ParentProcessId 4956, the interactive PowerShell session from which the commands were run.

7.3 Timestamps

Sysmon writes UtcTime in UTC, and Security events store TimeCreated in UTC. Event Viewer converts these to local time for display. For example, Event 1 shows UtcTime 06:11:43 and Logged 11:41:43 AM, a difference of 5 hours 30 minutes, matching IST. Analysts should always note which time base a source uses before comparing events.

8. Observations and limitations

- Sysmon ran with its default configuration, so RuleName is "-" and no custom filtering was applied. A tuned configuration would reduce noise in a real environment.
- The 4720 event in this report belongs to a second test account, TelemetryTest2; the 4722 event belongs to the first account, TelemetryTest.
- The Event 3 screenshot shows a single failed logon. Detecting brute force requires many failures over a short period.
- The account-creation command shows the test password in clear text in the PowerShell history and screenshot. This is acceptable for a throwaway lab account and is a good reminder that command lines are logged.
- The test accounts should be removed after the exercise, for example with net user TelemetryTest /delete.

9. Conclusion

Five distinct events were generated and located in the Windows logs: a process creation, a new user account, a failed logon, a PowerShell command and a Command Prompt command. Sysmon supplied the process view (Event ID 1) while the built-in Security log supplied account and logon activity (4720, 4722, 4625). For each event the timestamp, user and process were identified, and both log

files were located on disk under C:\Windows\System32\winevt\Logs. Together the two sources give a clear picture of who did what, when, and from which process.

Appendix A: Evidence index

Screenshots were stored in C:\LocalTelemetry\Screenshots, as listed below.

```
PS C:\Sysmon> Get-ChildItem "C:\LocalTelemetry\Screenshots"

Directory: C:\LocalTelemetry\Screenshots

Mode                LastWriteTime         Length Name
----                -
-a-----         10/4/2026  11:32 AM          50799 01_Audit_Configuration.png
-a-----         10/4/2026  11:38 AM          58978 02_Sysmon_installation .png
-a-----         10/4/2026  11:43 AM       200842 03_eventid.png
-a-----         10/4/2026  11:48 AM       308675 04_Event2_UserCreation.png
-a-----         10/4/2026  12:12 PM       389533 04_Event2_UserCreation4720.png
-a-----         10/4/2026  11:45 AM         54235 04_password.png
-a-----         10/4/2026  11:55 AM       429645 05_Event3_FailedLogin.png
-a-----         10/4/2026  12:02 PM       346057 06_Event4_PowerShell.png
-a-----         10/4/2026  12:30 PM       408600 07_Event5_CMD.png
-a-----         10/4/2026  12:32 PM         37498 08_Log_File_Locations.png
-a-----         10/4/2026  12:33 PM       159273 09_Sysmon_Log_Path.png
-a-----         10/4/2026  11:36 AM         27447 sysmon_install.png

PS C:\Sysmon>
```

Figure 13: Directory listing of the screenshot evidence folder.

File	Used in this report
01_Audit_Configuration.png	Section 3.1
sysmon_install.png	Section 3.2 (Sysmon package contents)
02_Sysmon_installation_.png	Section 3.2 (installation)
03_eventid.png	Section 5.1 (Event 1)
04_password.png	Section 5.2 (account creation command)
04_Event2_UserCreation.png	Section 5.2 (Event ID 4722)
04_Event2_UserCreation4720.png	Section 5.2 (Event ID 4720)
05_Event3_FailedLogin.png	Section 5.3 (Event 3)
06_Event4_PowerShell.png	Section 5.4 (Event 4)
07_Event5_CMD.png	Section 5.5 (Event 5)
08_Log_File_Locations.png	Section 6
09_Sysmon_Log_Path.png	Section 6
11_Evidence_Index.png	Appendix A